

SALEM, OR, USA

WMO: 726940

Lat: 44.905N

Lon: 123.001W

Elev: 63

StdP: 100.58

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.1	-2.3	-10.0	1.6	-1.6	-6.8	2.1	0.7	12.6	10.0	11.4	10.3	1.5	340	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	33.7	19.3	31.4	18.8	29.2	18.0	20.3	31.2	19.5	29.7	18.6	28.0	4.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	11.7	22.9	15.5	11.1	22.4	14.8	10.6	21.7	58.4	31.4	55.4	29.6	52.6	27.9	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.1	7.1	-7.5	37.8	2.2	2.0	-9.0	39.2	-10.3	40.4	-11.5	41.5	-13.1	43.0		
			DB													
			WB	-8.1	22.1	2.3	1.0	-9.8	22.8	-11.1	23.4	-12.4	23.9	-14.0	24.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	5.5	6.4	8.4	10.5	14.0	16.8	20.6	20.5	17.6	12.2	7.9	5.1
	DBStd	6.29	3.45	3.01	2.73	2.90	3.26	3.19	2.98	2.77	2.98	2.98	3.37	3.62
	HDD10.0	592	143	104	64	27	4	0	0	0	0	13	78	157
	HDD18.3	2464	398	334	307	237	141	66	10	9	49	191	312	411
	CDD10.0	1379	4	4	16	41	127	206	328	326	228	81	16	4
	CDD18.3	211	0	0	0	1	6	21	79	77	27	1	0	0
	CDH23.3	2823	0	0	0	22	137	339	1005	908	393	19	0	0
	CDH26.7	1127	0	0	0	3	38	127	440	378	138	3	0	0
Wind	WSAvg	2.8	3.0	3.0	3.2	3.0	2.7	2.8	2.8	2.5	2.4	2.5	3.0	3.1
Precipitation	PrecAvg	999	153	115	107	66	52	35	12	17	35	73	159	168
	PrecMax	1406	342	290	217	143	141	106	67	106	101	164	389	315
	PrecMin	603	6	19	22	16	1	7	0	0	0	3	29	32
	PrecStd	188	75	66	51	31	29	23	16	22	28	43	76	76
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	16.5	21.4	26.3	30.5	34.4	37.2	36.1	33.5	25.7	18.0	15.2
		MCWB	12.8	11.2	13.2	16.1	17.8	19.8	20.8	19.7	18.4	15.4	14.7	12.6
	2%	DB	13.3	14.4	18.1	22.3	27.2	30.2	33.8	33.2	30.3	22.6	15.8	13.2
		MCWB	10.9	10.3	11.6	14.2	16.9	18.4	19.8	19.1	17.5	14.3	13.2	11.4
	5%	DB	11.9	12.8	15.6	19.1	24.1	27.1	31.3	30.8	27.9	19.9	14.3	11.6
		MCWB	9.9	9.7	10.7	12.4	15.7	17.2	19.2	18.7	17.1	13.7	12.0	10.2
	10%	DB	10.6	11.4	13.7	16.4	21.3	24.4	28.9	28.4	25.2	17.6	12.9	10.1
		MCWB	9.0	8.8	9.9	11.2	14.3	16.0	18.3	18.0	16.3	13.2	10.9	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	13.0	14.1	16.8	19.3	20.6	22.0	21.1	20.2	17.7	15.7	13.3
		MCDB	14.7	14.6	18.9	24.8	28.1	32.2	34.4	33.3	29.9	22.0	17.5	14.5
	2%	WB	11.5	11.5	12.6	14.7	17.5	19.0	20.6	20.1	18.9	15.8	13.9	11.6
		MCDB	12.7	13.2	16.3	21.2	25.3	28.9	31.5	31.1	27.6	19.7	15.3	12.8
	5%	WB	10.2	10.2	11.4	13.0	16.1	17.7	19.6	19.2	17.8	14.7	12.4	10.3
		MCDB	11.5	12.2	14.7	17.9	23.0	26.2	30.0	29.2	25.7	18.0	13.7	11.5
	10%	WB	9.1	9.0	10.4	11.7	14.8	16.5	18.5	18.3	16.8	13.8	11.3	8.9
		MCDB	10.5	11.0	13.2	15.6	20.7	23.5	28.1	27.4	23.8	16.8	12.6	10.1
Mean Daily Temperature Range	5% DB	MDBR	6.2	8.1	9.3	10.4	11.4	12.3	14.9	14.8	13.8	10.4	7.2	6.0
		MCDBR	7.3	10.4	13.6	16.3	17.7	18.0	19.2	18.9	18.9	15.1	8.1	6.4
		MCWBR	5.2	6.7	7.9	9.0	8.9	7.9	7.4	7.1	7.9	7.8	5.4	5.0
	5% WB	MCDBR	6.1	7.9	11.3	14.1	15.7	16.9	17.9	17.3	16.4	10.8	6.4	5.8
		MCWBR	4.9	5.6	7.0	8.2	8.3	7.8	7.4	7.0	7.3	6.3	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.316	0.316	0.326	0.355	0.365	0.358	0.350	0.353	0.343	0.330	0.324	0.322	
	taud	2.535	2.542	2.518	2.400	2.391	2.418	2.462	2.472	2.492	2.535	2.517	2.489	
	Ebn at Noon	807	878	913	907	904	909	912	898	881	846	786	753	
	Edh at Noon	67	78	91	111	116	113	107	102	93	78	67	64	
All-Sky Solar Radiation	RadAvg	1.25	2.07	3.02	4.36	5.55	6.19	7.07	6.05	4.52	2.65	1.43	1.01	
	RadStd	0.20	0.29	0.39	0.39	0.55	0.56	0.45	0.38	0.38	0.31	0.19	0.15	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+38
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	+22

Nomenclature: See separate page